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Neural Control of Exercising Muscle

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In Closing

Josh Harding retired from the National Hockey League (NHL) in 2015 after an 8-year career as a goalie, posting 60 NHL wins. While warming up for a game, Harding felt a tweak in his neck, followed by dizziness, black spots in front of his eyes, and numbness in his right leg. In December of 2012, just before the 2012-2013 NHL season began, Harding learned he had multiple sclerosis (MS), a disease that attacks the central nervous system and causes a loss of balance and coordination, blurred vision, dehydration, muscle spasms, and weakness. His team, the Minnesota Wild, was made aware of the diagnosis, and Harding eventually made the news public. Despite the challenges that this diagnosis imposes on an NHL goaltender, Harding was determined to continue to play, and play well he did for a while. However, after playing two periods for the minor-league Iowa Wild in 2014, Harding was taken by ambulance to the hospital suffering from severe dehydration, a common effect of MS. In his first full season after being diagnosed, he played 29 games with an 18-7-3 record, a 1.65 goals against average, and a 0.933 save percentage. Harding received the Bill Masterson Memorial Trophy in recognition of his perseverance and dedication to the game. Having found the correct combination of medication and a sleep schedule that works, he now serves as a high school goaltender coach while raising three young children.

All functions within the human body are influenced in some way by the nervous system. Nerves are the wiring through which electrical impulses are sent to and received from virtually all tissues of the body. The brain acts as a central computer, integrating incoming information, selecting an appropriate response, and then signaling the involved organs and tissues to take appropriate action. Thus, the nervous system forms a vital network, allowing communication, coordination, and interaction of the various tissues and systems in the body as well as between the body and the external environment.

The nervous system is one of the body's most complex systems. Because this book is primarily concerned with neural control of muscle contraction and voluntary movement, we will limit our coverage of this complex system. We first review the structure and function of the nervous system and then focus on specific topics relevant to sport and exercise.

Before we examine the intricate details of the nervous system, it is important to look at how the nervous system is organized and how

that organization functions to integrate and control movement. The nervous system is commonly divided into two parts: the **central nervous system (CNS)** and the **peripheral nervous system (PNS)**. The CNS is composed of the brain and spinal cord, while the PNS is further divided into **sensory (afferent) nerves** and **effector (efferent) nerves**. Sensory nerves are responsible for informing the CNS about what is going on within and outside the body. Efferent nerves are responsible for sending information from the CNS to the various tissues, organs, and systems of the body in response to the signals coming in from the sensory division. The term **motor neuron (motor nerve)** classically applies to neurons that project their axons outside the CNS to directly or indirectly control muscles. The efferent nervous system is composed of two parts, the autonomic nervous system and the somatic nervous system. [Figure 3.1](#) provides a schematic of these relationships. More detail concerning each of these individual units of the nervous system is presented later in this chapter.

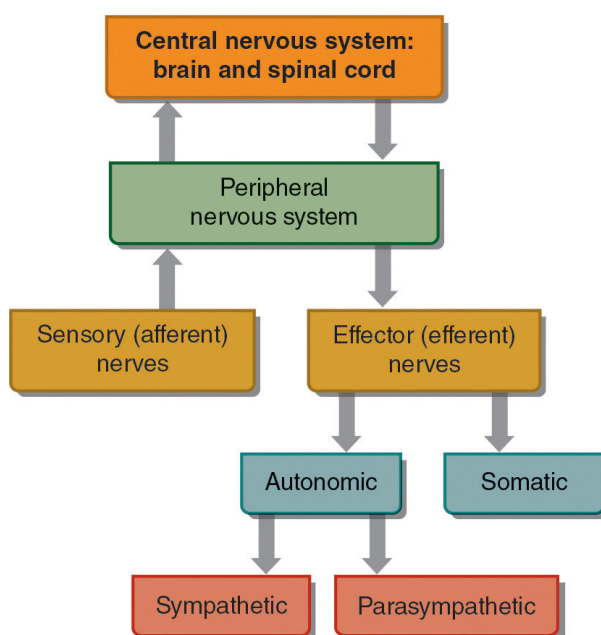


FIGURE 3.1 Organization of the nervous system.

Structure and Function of the Nervous System

The **neuron** is the basic structural unit of the nervous system. We first review the anatomy of the neuron and then look at how it

functions—allowing electrical impulses to be transmitted throughout the body.

Neuron

Individual nerve fibers (nerve cells), depicted in [figure 3.2](#), are called neurons. A typical neuron is composed of three regions:

- The cell body, or soma
- The dendrites
- The axon

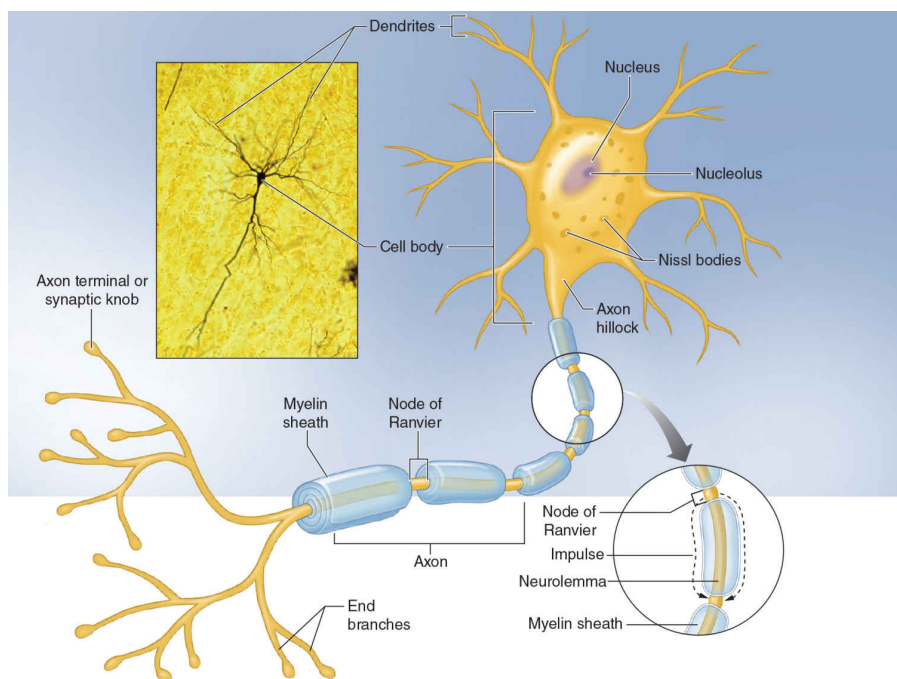


FIGURE 3.2 A drawing and photomicrograph (inset) of a neuron and its structure.

The cell body contains the nucleus. Radiating out from the cell body are the two cell processes: dendrites and the axon. On the side toward the axon, the cell body tapers into a cone-shaped region known as the **axon hillock**. The axon hillock has an important role in impulse conduction, as discussed later.

Most neurons contain only one axon but many dendrites. Dendrites are the neuron's receivers. Most impulses, or action potentials, that enter the neuron from sensory stimuli or from adjacent neurons typically enter the neuron via the dendrites. These processes then carry the impulses toward the cell body.

The axon is the neuron's transmitter and conducts impulses away from the cell body. Near its end, an axon splits into numerous **end branches**. The tips of these branches are dilated into tiny bulbs known as **axon terminals** or synaptic knobs. These terminals or knobs house numerous vesicles (sacs) filled with chemicals known as **neurotransmitters** that are used for communication between a neuron and another cell. (This is discussed in more detail later in the chapter.) The structure of the neuron allows nerve impulses to enter the neuron through the dendrites, and to a lesser extent through the cell body, and to travel through the cell body and axon hillock, down the axon, and out through the end branches to the axon terminals. We next look in more detail at how this happens, including how these impulses travel from one neuron to another and from a somatic motor neuron to muscle fibers.

Nerve Impulse

Neurons are referred to as *excitable tissue* because they can respond to various types of stimuli and convert those messages to an electrical signal called a nerve impulse. A **nerve impulse** arises when a stimulus is strong enough to substantially change the normal electrical charge of the neuron. That signal then moves along the neuron down the axon and toward an end organ, such as another neuron or a group of muscle fibers. A useful analogy is between the nerve impulse traveling through a neuron and electricity traveling through the electrical wires in a home. This section describes how the electrical impulse is generated and how it travels through a neuron.

Resting Membrane Potential

The cell membrane of a typical neuron at rest has a negative electrical potential of about -70 mV. This means that if one were to insert a voltmeter probe inside the cell, the electrical charges found there and the charges found outside the cell would differ by 70 mV, and the inside would be negative relative to the outside. This electrical potential difference is known as the **resting membrane potential (RMP)**. It is caused by an uneven separation of charged ions across the membrane. When the charges across the membrane differ, the membrane is said to be polarized.

The neuron has a high concentration of potassium ions (K^+) on the inside of the membrane and a high concentration of sodium ions (Na^+) on the outside. The imbalance in the number of ions inside and outside the cell causes the RMP. This imbalance is maintained in two ways. First, the cell membrane is much more permeable to K^+ than to Na^+ , so the K^+ can move more freely. Because ions tend to move to establish equilibrium, some of the K^+ will move to the area where they are less concentrated, outside the cell. The Na^+ cannot move to the inside as easily. Second, **sodium–potassium pumps** in the neuron membrane, which contain Na^+ - K^+ adenosine triphosphatase (Na^+ - K^+ -ATPase), maintain the imbalance on each side of the membrane by actively transporting potassium ions in and sodium ions out. The sodium–potassium pump moves three Na^+ out of the cell for each two K^+ it brings in. The end result is that more positively charged ions are outside the cell than inside, creating the potential difference across the membrane. Maintenance of a constant RMP of about -70 mV is primarily a function of the sodium–potassium pump.

Depolarization and Hyperpolarization

If the inside of the cell becomes less negative relative to the outside, the potential difference across the membrane decreases. The membrane will be less polarized. When this happens, the membrane is said to be depolarized. Thus, **depolarization** occurs any time the charge difference becomes more positive than the RMP of -70 mV, moving closer to zero. This typically results from a change in the membrane's Na^+ permeability.

The opposite can also occur. If the charge difference across the membrane increases, moving from the RMP to an even more negative value, then the membrane becomes more polarized. This is known as **hyperpolarization**. Changes in the membrane potential control the signals used to receive, transmit, and integrate information within and between cells. These signals are of two types, graded potentials and action potentials. Both are electrical currents created by the movement of ions.

Graded Potentials

Graded potentials are localized changes in the membrane potential, either depolarization or hyperpolarization. The membrane

contains ion channels with gates that act as doorways into and out of the neuron. These gates are usually closed, preventing a large number of ions from flowing into and out of the membrane—that is, above and beyond the constant movement of Na^+ and K^+ that maintains the RMP. However, with potent enough stimulation, the gates open, allowing more ions to move from the outside to the inside or vice versa. This ion flow alters the charge separation, changing the polarization of the membrane.

Graded potentials are triggered by a change in the neuron's local environment. Depending on the location and type of neuron involved, the ion gates may open in response to the transmission of an impulse from another neuron or in response to sensory stimuli such as changes in chemical concentrations, temperature, or pressure.

Recall that most neuron receptors are located on the dendrites (although some are on the cell body), yet the impulse is always transmitted from the axon terminals at the opposite end of the cell. For a neuron to transmit an impulse, the impulse must travel almost the entire length of the neuron. Although a graded potential may result in depolarization of the entire cell membrane, it is usually just a local event such that the depolarization does not spread very far along the neuron. To travel the full distance, an impulse must be sufficiently strong to generate an action potential.

Action Potentials

An action potential is a rapid and substantial depolarization of the neuron's membrane. It usually lasts only about 1 ms. Typically, the membrane potential changes from the RMP of about -70 mV to a value of about $+30$ mV and then rapidly returns to its resting value. This is illustrated in [figure 3.3](#). How does this marked change in membrane potential occur?

All action potentials begin as graded potentials. When enough stimulation occurs to cause a depolarization of at least 15 to 20 mV, an action potential results. In other words, if the membrane depolarizes from the RMP of -70 mV to a value of -50 to -55 mV, an action potential will occur. The membrane voltage at which a graded potential becomes an action potential is called the depolarization **threshold**. Any depolarization that does not attain the

threshold will not result in an action potential. For example, if the membrane potential changes from the RMP of -70 mV to -60 mV, the change is only 10 mV and does not reach the threshold; thus, no action potential occurs. But any time depolarization reaches or exceeds the threshold, an action potential will result. This is commonly referred to as the *all-or-none principle*.

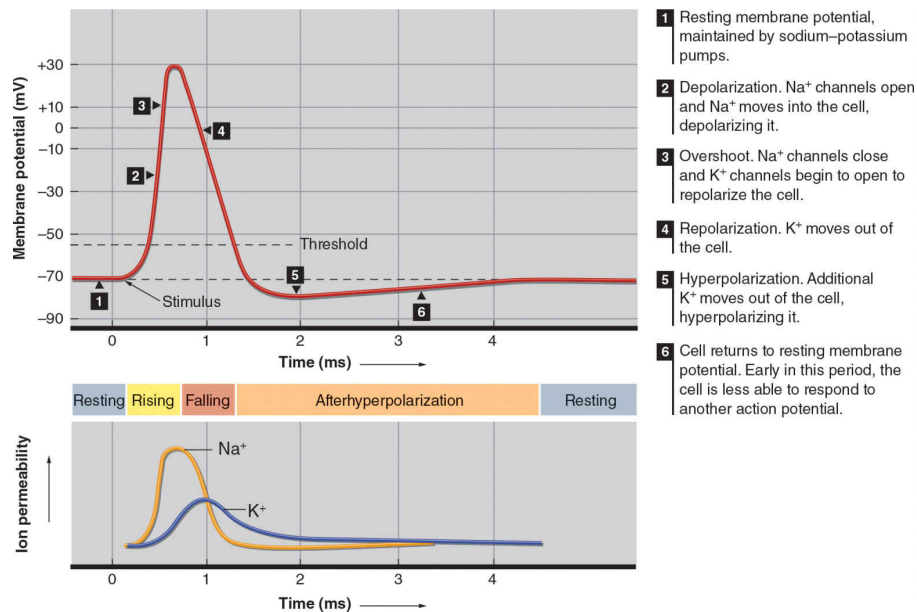


FIGURE 3.3 Voltage and ion permeability changes during an action potential.



When a segment of an axon's sodium gates is open and it is in the process of generating an action potential, it is unable to respond to another stimulus. This is referred to as the *absolute refractory period*. When the sodium gates are closed, the potassium gates are open, and repolarization is occurring, that segment of the axon can potentially respond to a new stimulus, but the stimulus must be of substantially greater magnitude to evoke an action potential. This is referred to as the *relative refractory period*.

Propagation of the Action Potential

Now that we understand how a neural impulse, in the form of an action potential, is generated, we can look at how the impulse is propagated—that is, how it travels through the neuron. Two characteristics of the neuron determine how quickly an impulse can pass along the axon: myelination and diameter.

Myelination The axons of many neurons, especially large neurons, are myelinated, meaning that they are covered with a sheath formed by myelin, a fatty substance that insulates the cell membrane. This **myelin sheath** (see [figure 3.2](#)) is formed by specialized cells called Schwann cells.

The myelin sheath is not continuous. As it spans the length of the axon, the myelin sheath has gaps between adjacent Schwann cells, leaving the axon uninsulated at those points. These gaps are referred to as *nodes of Ranvier* (see [figure 3.2](#)). The action potential appears to jump from one node to the next as it traverses a myelinated fiber. This is referred to as **saltatory conduction**, a much faster type of conduction than occurs in unmyelinated fibers.

Myelination of peripheral motor neurons occurs over the first several years of life, partly explaining why children need time to develop coordinated movement. Individuals affected by certain neurological diseases (such as multiple sclerosis, as discussed in our chapter opening) experience degeneration of the myelin sheath and a subsequent loss of coordination.

Diameter of the Neuron The velocity of nerve impulse transmission is also determined by the neuron's size. Neurons of larger diameter conduct nerve impulses faster than neurons of smaller diameter because larger neurons present less resistance to local current flow.

In Review

- Neurons are excitable tissues because they have the ability to respond to various types of stimuli and convert them to an electrical signal or nerve impulse.
- A neuron's RMP of about -70 mV results from the unequal separation of positively charged sodium and potassium ions, with more potassium inside the membrane and more sodium on the outside.
- The RMP is maintained by actions of the sodium–potassium pump, coupled with low sodium permeability and high potassium permeability of the neuron

membrane.

- Any change that makes the membrane potential less negative results in depolarization. Any change making this potential more negative is a hyperpolarization. These changes occur when ion gates in the membrane open, permitting more ions to move across the membrane.
 - If the membrane is depolarized by 15 to 20 mV, the depolarization threshold is reached and an action potential results. Action potentials are not generated if the threshold is not met.
 - In myelinated neurons, the impulse travels through the axon by jumping between nodes of Ranvier (gaps between the cells that form the myelin sheath). This process, saltatory conduction, results in nerve transmission rates 5 to 50 times faster than in unmyelinated fibers of the same size. Impulses also travel faster in neurons of larger diameter.
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Synapse

For a neuron to communicate with another neuron, an action potential must occur and travel along the first neuron, ultimately reaching its axon terminals. How does the action potential then move from the neuron in which it starts to another neuron to continue transmitting the electrical signal?

Neurons communicate with each other across junctions called synapses. A **synapse** is the site of action potential transmission from the axon terminals of one neuron to the dendrites or soma of another. There are both chemical and mechanical synapses, but the most common type is the chemical synapse, which is our focus. It is important to note that the signal that is transmitted from one neuron to another changes from electrical to chemical, then back to electrical.

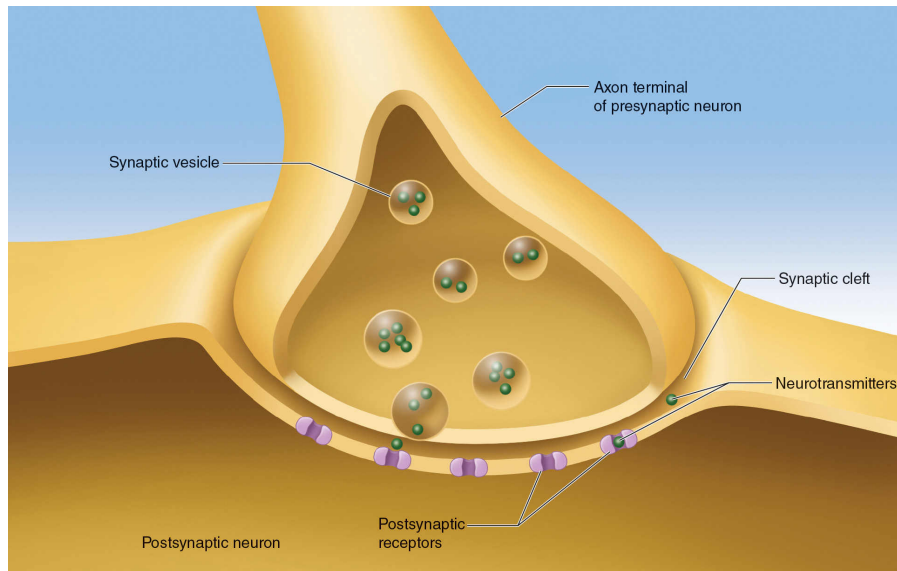


FIGURE 3.4 A chemical synapse between two neurons, showing the synaptic vesicles containing neurotransmitter molecules.

As seen in [figure 3.4](#), a synapse between two neurons includes

- the axon terminals of the neuron sending the action potential,
- receptors on the neuron receiving the action potential, and
- the space between these structures.

The neuron sending the action potential across the synapse is called the presynaptic neuron, so axon terminals are presynaptic terminals. Similarly, the neuron receiving the action potential on the opposite side of the synapse is called the postsynaptic neuron, and it has postsynaptic receptors. The axon terminals and postsynaptic receptors are not physically in contact with each other. A narrow gap, the synaptic cleft, separates them.

The action potential can be transmitted across a synapse in only one direction: from the axon terminal of the presynaptic neuron to the postsynaptic receptors, about 80% to 95% of which are on the dendrites of the postsynaptic neuron. The remaining 5% to 20% of the postsynaptic receptors are adjacent to the cell body instead of being located on the dendrites. Why can the action potential go in only one direction?

The presynaptic terminals of the axon contain a large number of saclike structures called synaptic (or storage) vesicles. These vesicles contain a variety of chemical compounds called

neurotransmitters because they function to transmit the neural signal to the next neuron. When the impulse reaches the presynaptic axon terminals, the synaptic vesicles respond by releasing the neurotransmitters into the synaptic cleft. These neurotransmitters then diffuse across the synaptic cleft to the postsynaptic neuron's receptors. Each neurotransmitter then binds to its specialized postsynaptic receptors. When sufficient binding occurs, a series of graded depolarizations occurs. If the depolarization reaches the threshold, an action potential occurs, and the impulse has been transmitted successfully to the next neuron. Depolarization of the second nerve depends on both the amount of neurotransmitter released and the number of available receptor binding sites on the postsynaptic neuron.

Neuromuscular Junction

Recall from [chapter 1](#) that a single α -motor neuron and all of the muscle fibers it innervates is called a motor unit. Whereas neurons communicate with other neurons at synapses, an α -motor neuron communicates with its muscle fibers at a site known as a **neuromuscular junction**, which functions in essentially the same manner as a synapse. In fact, the proximal part of the neuromuscular junction is the same: It starts with the axon terminals of the motor neuron, which release neurotransmitters into the space between the motor nerve and the muscle fiber in response to an action potential. However, in the neuromuscular junction, the axon terminals protrude into motor end plates, which are invaginated (folded to form cavities) segments on the plasmalemma of the muscle fiber (see [figure 3.5](#)).

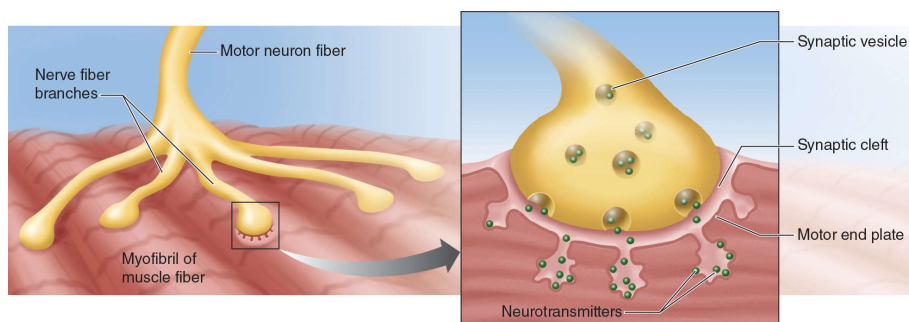


FIGURE 3.5 The neuromuscular junction, illustrating the interaction between the α -motor neuron and the plasmalemma of a single muscle fiber.

Neurotransmitters—primarily acetylcholine (ACh)—released from the α -motor neuron axon terminals diffuse across the synaptic cleft and bind to receptors on the muscle fiber's plasmalemma. This binding typically causes depolarization by opening sodium ion channels, allowing more sodium to enter the muscle fiber. Again, if the depolarization reaches the threshold, an action potential is formed. It spreads across the plasmalemma into the T-tubules, initiating muscle fiber contraction. As in the neuron, the plasmalemma, once depolarized, must undergo repolarization. During the period of repolarization, the sodium gates are closed and the potassium gates are open; thus, like the neuron, the muscle fiber is unable to respond to any further stimulation during this refractory period. Once the RMP of the muscle fiber is restored, the fiber can respond to another stimulus. Thus, the refractory period limits the motor unit's firing frequency.

Exercise training induces changes not only in skeletal muscle, but also at the neuromuscular junction (NMJ) to increase presynaptic release of, and sensitivity of the muscle cell to, acetylcholine. These changes occur through a number of different cellular signaling mechanisms; however, many of the changes induced by training share a common signaling molecule, the *peroxisome proliferator-activated receptor- γ coactivator 1 α* (PGC-1 α). PGC-1 α contributes to the remodeling of the NMJ in several ways. First, PGC-1 α induces adaptations in the motor neuron itself by increasing branching of the presynaptic terminal motor neuron and increasing the number of presynaptic vesicles containing acetylcholine. Second, PGC-1 α increases the number of acetylcholine receptors on the cell membrane, thus amplifying the effects for a given amount of acetylcholine released from the motor neuron.⁴ Finally, PGC-1 α is involved in decreasing the size of the motor end plate (i.e., fewer fibers per motor unit) on glycolytic fibers, making them similar to more oxidative fibers.

Muscular fatigue (discussed in detail in [chapter 5](#)) is a complex phenomenon, with many possible contributing factors. One mechanism that may contribute to muscle fatigue is a decline in signal transmission through the NMJ. Prior exercise can decrease

motor nerve outflow and neuromuscular transmission rates,³ which leads to decreased force production.

Now we know how the impulse is transmitted from nerve to nerve or nerve to muscle. But to understand what happens once the impulse is transmitted, we must first examine the chemical signaling molecules, the neurotransmitters, that accomplish this signal transmission.

Neurotransmitters

More than 50 neurotransmitters have been positively identified or are suspected to be potential candidates. These can be categorized as either (a) small-molecule, rapid-acting neurotransmitters or (b) neuropeptide, slow-acting neurotransmitters. The small-molecule, rapid-acting transmitters, which are responsible for most neural transmissions, are our main focus.

RESEARCH PERSPECTIVE 3.1

Motor Units Adapt to High-Intensity Interval Training

High-intensity interval training (HIIT, discussed in [chapter 11](#)) is a mode of physical activity that involves brief, intermittent bursts of vigorous activity interspersed with periods of low-intensity exercise or rest. An individual can reap the same cardiovascular and musculoskeletal benefits from exercise training using HIIT in far less time compared to traditional long-duration endurance training (END). Because HIIT is now a common alternative to END and exercise training changes the neural control of muscle function, it is important to systematically evaluate HIIT-induced neuromuscular adaptations.

High-density surface electromyography (EMG) is a relatively new technological advancement that allows for both the simultaneous assessment of several motor units over a wide range of forces and the ability to track the same motor units during different sessions over a long period of time (like during exercise training). Recording motor unit activity and function allows investigators to assess the way the nervous system controls muscle force. Researchers recently evaluated differences in the neuromuscular adaptations to HIIT and END using this technique.⁶ Two weeks of HIIT and END elicited similar improvements in cardiorespiratory fitness, but there were distinct adjustments in motor unit behavior with the two types of training. HIIT increased both maximum force production and motor unit discharge. In contrast, END did not influence motor unit firing. These findings suggest that HIIT and END have very different effects on motor unit function and provide

important new information regarding exercise training–induced neuromuscular adaptations. This study was also the first to demonstrate training-induced changes in motor unit discharge rate by tracking the *same* individual motor units before and after training. This innovative methodology will likely continue to broaden our understanding of neural adaptations to exercise training.

Acetylcholine and norepinephrine are the two major neurotransmitters involved in regulating multiple physiological responses to exercise. **Acetylcholine** is the primary neurotransmitter for the motor neurons that innervate skeletal muscle as well as for most parasympathetic autonomic neurons. It is generally an excitatory neurotransmitter in the somatic nervous system but can have inhibitory effects at some parasympathetic nerve endings, such as in the heart. **Norepinephrine** is the neurotransmitter for most sympathetic autonomic neurons, and it too can be either excitatory or inhibitory, depending on the receptors involved. Nerves that primarily release norepinephrine are called **adrenergic**, and those that have acetylcholine as their primary neurotransmitter are termed **cholinergic**. Two major subtypes of cholinergic receptors are muscarinic and nicotinic, with the former involved in motor nerve transmission. The sympathetic and parasympathetic branches of the autonomic nervous systems are discussed later in this chapter.

Once the neurotransmitter binds to the postsynaptic receptor, the nerve impulse has been successfully transmitted. The neurotransmitter then (1) is degraded by enzymes, (2) is actively transported back into the presynaptic terminals for reuse, or (3) diffuses away from the synapse.

In Review

- Neurons communicate with each other across synapses composed of the axon terminals of the presynaptic neuron, the postsynaptic receptors on the dendrite or cell body of the postsynaptic neuron, and the synaptic cleft between the two neurons.
- A nerve impulse causes neurotransmitters to be released from the presynaptic axon terminal into the synaptic cleft.

- Neurotransmitters diffuse across the cleft and bind to the postsynaptic receptors.
 - Once sufficient neurotransmitters are bound, the impulse is successfully transmitted and the neurotransmitter is then destroyed by enzymes, is removed by reuptake into the presynaptic terminal for future use, or diffuses away from the synapse.
 - Neurotransmitter binding at the postsynaptic receptors opens ion gates in the given membrane and can cause depolarization (excitation) or hyperpolarization (inhibition), depending on the specific neurotransmitter and the receptors to which it binds.
 - Neurons communicate with muscle fibers at neuromuscular junctions. A neuromuscular junction involves presynaptic axon terminals, the synaptic cleft, and motor end-plate receptors on the plasmalemma of the muscle fiber and functions much like a neural synapse.
 - The neurotransmitters most important in regulating exercise responses are acetylcholine in the somatic nervous system and norepinephrine in the autonomic nervous system.
 - Receptors on the motor end plates of the neuromuscular junction are a special subtype of cholinergic receptors called muscarinic receptors. They bind the primary neurotransmitter involved in excitation of muscle fibers, acetylcholine.
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Postsynaptic Response

Once the neurotransmitter binds to the receptors, the chemical signal that traversed the synaptic cleft once again becomes an electrical signal. The binding causes a graded potential in the postsynaptic membrane. An incoming impulse may be either excitatory or inhibitory. An excitatory impulse causes depolarization, known as an **excitatory postsynaptic potential (EPSP)**. An inhibitory impulse causes a hyperpolarization, known as an **inhibitory postsynaptic potential (IPSP)**.

The discharge of a single presynaptic terminal generally changes the postsynaptic potential less than 1 mV. Clearly this is not sufficient to generate an action potential, because reaching the threshold requires a change of at least 15 mV. But when a neuron transmits an impulse, several presynaptic terminals typically release their neurotransmitters so that they can diffuse to the postsynaptic receptors. Also, presynaptic terminals from numerous axons can

converge on the dendrites and cell body of a single neuron. When multiple presynaptic terminals discharge at the same time, or when only a few fire in rapid succession, more neurotransmitter is released. With an excitatory neurotransmitter, the more that is bound, the greater the EPSP and the more likely it is that an action potential will result.

In Review

- Excitatory postsynaptic potentials are graded depolarizations of the postsynaptic membrane; IPSPs are hyperpolarizations of that membrane.
- A single presynaptic terminal cannot generate enough of a depolarization to fire an action potential. Multiple signals are needed. These may come from numerous neurons or from a single neuron when numerous axon terminals release neurotransmitters repeatedly and rapidly.
- The axon hillock keeps a running total of all EPSPs and IPSPs. When their sum meets or exceeds the threshold for depolarization, an action potential occurs. This process of accumulating incoming signals is known as summation.
- *Summation* refers to the cumulative effect of all individual graded potentials as processed by the axon hillock. Once the sum of all individual graded potentials meets or exceeds the depolarization threshold, an action potential occurs.

Triggering an action potential at the postsynaptic neuron depends on the combined effects of all incoming impulses from these various presynaptic terminals. A number of impulses are needed to cause sufficient depolarization to generate an action potential. Specifically, the sum of all changes in the membrane potential must equal or exceed the threshold. This accumulation of the individual impulses' effects is called summation.

RESEARCH PERSPECTIVE 3.2

Aging Reduces Rapid Strength

By 2030, it is anticipated that older adults (>65 years) will make up 20% of the total population. Unfortunately, a large percentage of older adults experience functional limitations in their activities of daily living, and one out of three older adults experiences a fall each year. Accidental falls often cause an accelerated deterioration in overall health and impart a significant

economic burden on society. Alterations in neuromuscular function have been suggested to contribute to the increased fall risk in older adults.

Although reduced maximal muscle strength is a well-understood characteristic of aging, recent studies have demonstrated that *rapid strength* (the rate of torque development, or RTD) actually decreases at a greater rate than maximal strength. Furthermore, RTD measured within the initial 200 ms from the onset of muscle contraction is more functionally relevant than the peak torque that can be produced by the muscle. Despite this knowledge, until recently, no research has specifically targeted the neural and muscle-specific factors that contribute to the reductions in RTD with aging. This is a clinically relevant topic, since this information may help identify strategies to slow age-related reductions in function, thus reducing the risk of fall-related injuries.

A group of researchers recently sought to determine the mechanisms of age-related reductions in RTD.² Young (20 years old) and older men (70 years old) participated in a study that involved ultrasound assessments of muscle properties and measurements of muscle strength during early (the first 50 ms) and late (100 to 200 ms) intervals following the onset of muscle contraction. RTD was reduced in the older men during the late interval of contraction, but surprisingly there were no differences in RTD between young and older men during the early interval of contraction. This suggests that older men have similar initial muscle activation but are unable to sustain the same rates of muscle activation during later muscle contraction. Poorer muscle quality and reductions in pennation angle also contribute to age-related reductions in RTD, likely because they affect muscle fiber shortening and fiber rotation. These age-related alterations in neuromuscular function combine to reduce rapid muscle strength, significantly decreasing neuromuscular function and contributing to falls in older adults.

For summation, the postsynaptic neuron must keep a running total of the neuron's responses, both EPSPs and IPSPs, to all incoming impulses. This task is done at the axon hillock, which lies on the axon just past the cell body. Only when the sum of all individual graded potentials meets or exceeds the threshold can an action potential occur.

Individual neurons are grouped together into bundles. In the CNS (brain and spinal cord), these bundles are referred to as tracts, or pathways. Neuron bundles in the PNS are referred to simply as nerves.

Central Nervous System

To comprehend how even the most basic stimulus can cause muscle activity, we next consider the complexity of the CNS. The CNS comprises more than 100 billion neurons. In this section, we present an overview of the components of the CNS and their functions.

Brain

The brain is a highly complex organ composed of numerous specialized areas. For our purposes, we subdivide it into the four major regions illustrated in [figure 3.6](#): the cerebrum, diencephalon, cerebellum, and brain stem.

Cerebrum

The cerebrum is composed of the right and left cerebral hemispheres. These are connected to each other by the corpus callosum, fiber bundles (tracts) that allow the two hemispheres to communicate with each other. The cerebral cortex forms the outer portion of the cerebral hemispheres and has been referred to as the site of the mind and intellect. It is also called the gray matter, which simply reflects its distinctive color resulting from lack of myelin on the neurons located in this area. The cerebral cortex is the conscious brain. It allows people to think, be aware of sensory stimuli, and voluntarily control their movements.

The cerebrum consists of five lobes—four outer lobes and the central insular lobe—having the following general functions (see [figure 3.6](#)):

- Frontal lobe: general intellect and motor control
- Temporal lobe: auditory input and interpretation
- Parietal lobe: general sensory input and interpretation
- Occipital lobe: visual input and interpretation
- Insular lobe: diverse functions usually linked to emotion and self-perception

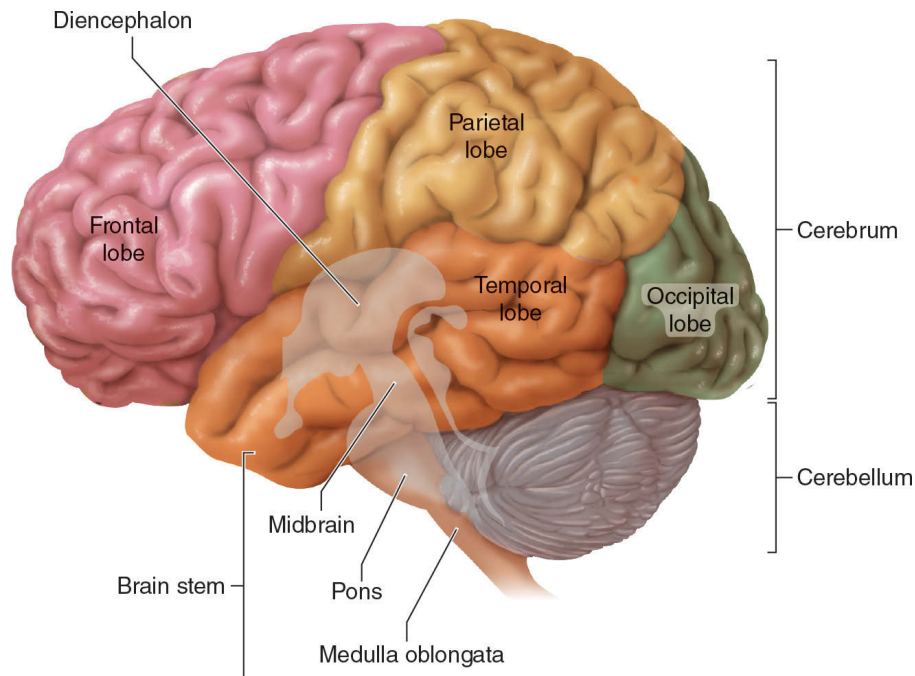


FIGURE 3.6 Four major regions of the brain and four outer lobes of the cerebrum. (Note that the insular lobe is not shown because it is folded deep within the cerebrum between the temporal lobe and the frontal lobe.)

The three areas in the cerebrum that are of primary concern to exercise physiology are the primary motor cortex, located in the frontal lobe; the basal ganglia, located in the white matter below the cerebral cortex; and the primary sensory cortex, located in the parietal lobe. In this section, the focus is on the primary motor cortex and basal ganglia, which work to control and coordinate movement.

Primary Motor Cortex The primary motor cortex is responsible for the control of fine and discrete muscle movements. It is located in the frontal lobe, specifically within the precentral gyrus. Neurons here, known as *pyramidal cells*, let us consciously control movement of skeletal muscles. Think of the primary motor cortex as the part of the brain where decisions are made about what movement one wants to make. For example, in baseball, if a player is in the batter's box waiting for the next pitch, the decision to swing the bat is made in the primary motor cortex, where the entire body is carefully mapped out. The areas that require the finest motor control have a greater representation in the motor cortex; thus, more neural control is provided to them.

The cell bodies of the pyramidal cells are housed in the primary motor cortex, and their axons form the extrapyramidal tracts. These are also known as the corticospinal tracts because the nerve processes extend from the cerebral cortex down to the spinal cord. These tracts provide the major voluntary control of skeletal muscles.

In addition to the primary motor cortex, there is a premotor cortex just anterior to the precentral gyrus in the frontal lobe. Learned motor skills of a repetitious or patterned nature are stored here. This region can be thought of as the memory bank for skilled motor activities.⁵

Basal Ganglia The basal ganglia (nuclei) are not part of the cerebral cortex. Rather, they are in the cerebral white matter, deep in the cortex. These ganglia are clusters of nerve cell bodies. The complex functions of the basal ganglia are not well understood, but the ganglia are known to be important in initiating movements of a sustained and repetitive nature (such as arm swinging during walking), and thus they control complex movements such as walking and running. These cells also are involved in maintaining posture and muscle tone.

Diencephalon

The region of the brain known as the diencephalon (see [figure 3.6](#)) contains the thalamus and the hypothalamus. The thalamus is an important sensory integration center. All sensory input (except smell) enters the thalamus and is relayed to the appropriate area of the cortex. The thalamus regulates what sensory input reaches the conscious brain and thus is very important for motor control.

The hypothalamus, directly below the thalamus, is responsible for maintaining homeostasis by regulating almost all processes that affect the body's internal environment. Neural centers here assist in the regulation of most physiological systems, including

- blood pressure, heart rate, and contractility;
- respiration;
- digestion;
- body temperature;
- thirst and fluid balance;
- neuroendocrine control;

- appetite and food intake; and
- sleep–wake cycles.

Cerebellum

The cerebellum is located behind the brain stem. It is connected to numerous parts of the brain and has a crucial role in *coordinating* movement.

The cerebellum is crucial to the control of all rapid and complex muscular activities. It helps coordinate the timing of motor activities and the rapid progression from one movement to the next by monitoring and making corrective adjustments in the motor activities that are elicited by other parts of the brain. The cerebellum assists the functions of both the primary motor cortex and the basal ganglia. It facilitates movement patterns by smoothing out the movement, which would otherwise be jerky and uncontrolled.

The cerebellum acts as an integration system, comparing the programmed or intended activity with the actual changes occurring in the body and then initiating corrective adjustments through the motor system. It receives information from the cerebrum and other parts of the brain and also from sensory receptors (proprioceptors) in the muscles and joints that keep the cerebellum informed about the body's current position. The cerebellum also receives visual and equilibrium input. Thus, it notes all incoming information about the exact tension and position of all muscles, joints, and tendons and the body's current position relative to its surroundings, then it determines the best plan of action to produce the desired movement.

After the primary motor cortex makes the decision to perform a movement, this decision is relayed to the cerebellum. The cerebellum notes the desired action and then compares the intended movement with the actual movement based on sensory feedback from the muscles and joints. If the action is different than planned, the cerebellum informs the higher centers of the discrepancy so corrective action can be initiated.

Brain Stem

The brain stem, composed of the midbrain, the pons, and the medulla oblongata (see [figure 3.6](#)), connects the brain and the spinal

cord. Sensory and motor neurons pass through the brain stem as they relay information in both directions between the brain and the spinal cord. This is the site of origin for 10 of the 12 pairs of cranial nerves. The brain stem also contains the major autonomic centers that control the respiratory and cardiovascular systems.

A specialized collection of neurons in the brain stem, known as the *reticular formation*, is influenced by, and has an influence on, nearly all areas of the CNS. These neurons help

- coordinate skeletal muscle function,
- maintain muscle tone,
- control cardiovascular and respiratory functions, and
- determine state of consciousness (arousal and sleep).

The brain has a pain control system located in the reticular formation, a group of nerve fibers in the brain stem. Opioid substances such as enkephalins and β -endorphin act on the opiate receptors in this region to help modulate pain. Research has demonstrated that exercise of long duration increases the concentrations of these substances. While this has been interpreted as the mechanism causing the “endorphin calm” or “runner’s high” experienced by some exercisers, the cause–effect association between endogenous opioids and these sensations has not been substantiated.

Spinal Cord

The lowest part of the brain stem, the medulla oblongata, is continuous with the spinal cord below it. The spinal cord is composed of tracts of nerve fibers that allow two-way conduction of nerve impulses. The sensory (afferent) fibers carry neural signals from sensory receptors, such as those in the skin, muscles, and joints, to the upper levels of the CNS. Motor (efferent) fibers from the brain and upper spinal cord transmit action potentials to end organs (e.g., muscles, glands).

In Review

- The CNS includes the brain and the spinal cord.

- The four major divisions of the brain are the cerebrum, the diencephalon, the cerebellum, and the brain stem.
 - The cerebral cortex is the conscious brain. The primary motor cortex, located in the frontal lobe, is the center of conscious motor control.
 - The basal ganglia, in the cerebral white matter, help initiate some movements (sustained and repetitive ones) and help control posture and muscle tone.
 - The diencephalon includes the thalamus, which receives all sensory input entering the brain, and the hypothalamus, which is a major control center for homeostasis.
 - The cerebellum, which is connected to numerous parts of the brain, is critical for coordinating movement. It is an integration center that decides how to best execute the desired movement, given the body's current position and the muscles' current status.
 - The brain stem is composed of the midbrain, the pons, and the medulla oblongata.
 - The spinal cord contains both sensory and motor fibers that transmit action potentials between the brain and the periphery.
-

Peripheral Nervous System

The PNS contains 43 pairs of nerves: 12 pairs of cranial nerves that connect with the brain and 31 pairs of spinal nerves that connect with the spinal cord. Cranial and spinal nerves directly supply the skeletal muscles. Functionally, the PNS has two major divisions: the sensory division and the motor division.

Sensory Division

The sensory division of the PNS carries sensory information toward the CNS. Sensory (afferent) neurons originate in such areas as blood vessels, internal organs, muscles and tendons, the skin, and sensory organs (taste, touch, smell, hearing, vision).

Sensory neurons in the PNS end in either the spinal cord or the brain and continuously convey information to the CNS concerning the body's constantly changing status, position, and internal and external environment. Sensory neurons within the CNS carry the sensory input to appropriate areas of the brain, where the

information can be processed and integrated with other incoming information.

The sensory division receives information from five primary types of receptors:

1. *Mechanoreceptors* that respond to mechanical forces such as pressure, touch, vibrations, or stretch
2. *Thermoreceptors* that respond to changes in temperature
3. *Nociceptors* that respond to painful stimuli
4. *Photoreceptors* that respond to electromagnetic radiation (light) to allow vision
5. *Chemoreceptors* that respond to chemical stimuli, such as from foods, odors, or changes in blood or tissue concentrations of substances like oxygen, carbon dioxide, glucose, and electrolytes

Virtually all of these receptors are important in exercise and sport. Special muscle and joint nerve endings are of many types and functions, and each type is sensitive to a specific stimulus. Here are some important examples:

- Free nerve endings detect crude touch, pressure, pain, heat, and cold. Thus, they function as mechanoreceptors, nociceptors, and thermoreceptors. These nerve endings are important for preventing injury during athletic performance.
- Joint kinesthetic receptors located in the joint capsules are sensitive to joint angles and rates of change in these angles. Thus, they sense the position and any movement of the joints.
- Muscle spindles sense muscle length and rate of change in length.
- Golgi tendon organs detect the tension applied by a muscle to its tendon, providing information about the strength of muscle contraction.

Muscle spindles and Golgi tendon organs are discussed in more detail later in this chapter.

Motor Division

The CNS transmits information to various parts of the body through the motor, or efferent, division of the PNS. Once the CNS has processed the information it receives from the sensory division, it determines how the body should respond to that input. From the brain and spinal cord, intricate networks of neurons go out to all parts of the body, providing detailed instructions to the target areas including—and central to exercise and sport physiology—muscles.

Autonomic Nervous System

The autonomic nervous system, often considered part of the motor division of the PNS, controls the body's involuntary internal functions. Some of these functions that are important to sport and activity are heart rate, blood pressure, blood distribution, and lung function.

The autonomic nervous system has two major divisions: the sympathetic nervous system and the parasympathetic nervous system. These originate from different sections of the spinal cord and from the base of the brain. The effects of the two systems are often antagonistic, but the systems always function together.

Sympathetic Nervous System

The sympathetic nervous system is sometimes called the fight-or-flight system: It prepares the body to face a crisis and sustains its function during the crisis. When fully engaged, the sympathetic nervous system can produce a massive discharge throughout the body, preparing it for action. A sudden loud noise, a life-threatening situation, or those last few seconds before the start of an athletic competition are examples of circumstances in which this massive sympathetic excitation may occur. The effects of sympathetic stimulation are important during exercise. To give a few examples:

- Heart rate and strength of cardiac contraction increase.
- Coronary vessels dilate, increasing the blood supply to the heart muscle to meet its increased demands.
- Peripheral vasodilation increases blood flow to active skeletal muscles.
- Vasoconstriction in most other tissues diverts blood away from them and to the active muscles.

- Blood pressure increases, allowing better perfusion of the muscles and improving the return of venous blood to the heart.
- Bronchodilation improves ventilation and effective gas exchange.
- Metabolic rate increases, reflecting the body's effort to meet the increased demands of physical activity.
- Mental activity increases, allowing better perception of sensory stimuli and more concentration on performance.
- Glucose is released from the liver into the blood as an energy source.
- Functions not directly needed at that time are slowed (e.g., renal function, digestion).

These basic alterations in bodily function facilitate motor responses, demonstrating the importance of the autonomic nervous system in preparing the body for and sustaining it during acute stress or physical activity.

Parasympathetic Nervous System

The parasympathetic nervous system can be thought of as the body's housekeeping system. It has a major role in carrying out such processes as digestion, urination, glandular secretion, and conservation of energy. This system is more active when one is calm and at rest. Its effects tend to oppose those of the sympathetic system. The parasympathetic division causes decreased heart rate, constriction of coronary vessels, and bronchoconstriction.

The various effects of the sympathetic and parasympathetic divisions of the autonomic nervous system are summarized in [table 3.1](#).

In Review

- The PNS contains 43 pairs of nerves: 12 cranial and 31 spinal.
- The PNS can be subdivided into the sensory and motor divisions. The motor division also includes the autonomic nervous system.
- The sensory division carries information from sensory receptors to the CNS. The motor division carries motor impulses from the CNS to the muscles and other

organs.

- The autonomic nervous system includes the sympathetic nervous system and the parasympathetic system. Although these systems often oppose each other, they always function together to create an appropriately balanced response.

Sensory-Motor Integration

Having discussed the components and divisions of the nervous system, we now discuss how a sensory stimulus gives rise to a motor response. How, for example, do the muscles in the hand know to pull one's finger away from a hot stove? When someone decides to run, how do the muscles in the legs coordinate while supporting weight and propelling the person forward? To accomplish these tasks, the sensory and motor systems must communicate with each other.

This process is called **sensory-motor integration**, and it is depicted in [figure 3.7](#). For the body to respond to sensory stimuli, the sensory and motor divisions of the nervous system must function together in the following sequence of events:

1. A sensory stimulus is received by sensory receptors (e.g., pinprick).
2. The sensory action potential is transmitted along sensory neurons to the CNS.
3. The CNS interprets the incoming sensory information and determines which response is most appropriate, or reflexively initiates a motor response.
4. The action potentials for the response are transmitted from the CNS along α -motor neurons.
5. The motor action potential is transmitted to a muscle, and the response occurs.

TABLE 3.1 Effects of the Sympathetic and Parasympathetic Nervous Systems on Various Organs

Target organ or system	Sympathetic effects	Parasympathetic effects
Heart muscle	Increases rate and force of contraction	Decreases rate of contraction
Heart: coronary blood vessels	Causes vasodilation	Causes

Lungs	Causes bronchodilation; mildly constricts blood vessels	vasoconstriction Causes bronchoconstriction Has little or no effect
Blood vessels	Increases blood pressure; causes vasoconstriction in abdominal viscera and skin to divert blood when necessary; causes vasodilation in the skeletal muscles and heart during exercise	
Liver	Stimulates glucose release	Has no effect
Cellular metabolism	Increases metabolic rate	Has no effect
Adipose tissue	Stimulates lipolysis ^a	Has no effect
Sweat glands	Increases sweating	Has no effect
Adrenal glands	Stimulates secretion of epinephrine and norepinephrine	Has no effect
Digestive system	Decreases activity of glands and muscles; constricts sphincters	Increases peristalsis and glandular secretion; relaxes sphincters
Kidney	Causes vasoconstriction; decreases urine formation	Has no effect

^aLipolysis is the process of breaking down triglyceride into its basic units to be used for energy.

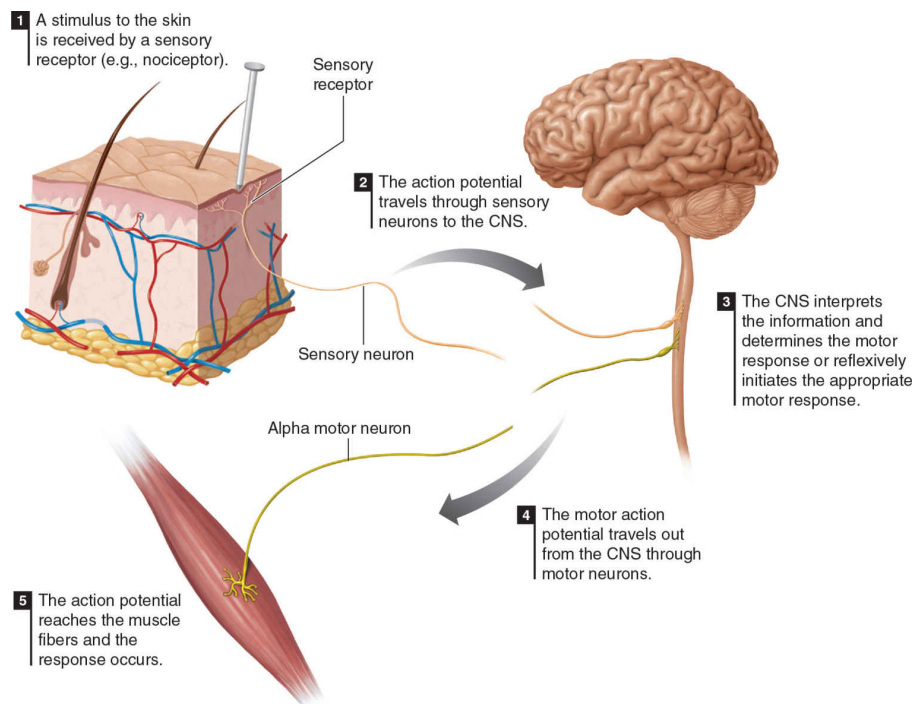


FIGURE 3.7 The sequence of events in sensory-motor integration.



Sensory Input

Recall that sensations and physiological status are detected by sensory receptors throughout the body. The action potentials resulting from sensory stimulation are transmitted via the sensory nerves to the spinal cord. When they reach the spinal cord, they can either trigger a local reflex at that level or travel to the upper regions of the spinal cord or to the brain. Sensory pathways to the brain can terminate in sensory areas of the brain stem, the cerebellum, the thalamus, or the cerebral cortex. An area in which the sensory impulses terminate is referred to as an integration center. This is where the sensory input is interpreted and linked to the motor system. [Figure 3.8](#) illustrates various sensory receptors and their nerve pathways back to the spinal cord and up into various areas of the brain. The integration centers vary in function:

- Sensory impulses that terminate in the spinal cord are integrated there. The response is typically a simple motor reflex, which is the simplest type of integration.
- Sensory signals that terminate in the lower brain stem result in subconscious motor reactions of a higher and more complex nature than simple spinal cord reflexes. Postural control during sitting, standing, or moving is an example of this level of sensory input.
- Sensory signals that terminate in the cerebellum also result in subconscious control of movement. The cerebellum appears to be the center of coordination, smoothing out movements by coordinating the actions of the various contracting muscle groups to perform the desired movement. Both fine and gross motor movements appear to be coordinated by the cerebellum in concert with the basal ganglia. Without the control exerted by the cerebellum, all movement would be uncontrolled and uncoordinated.

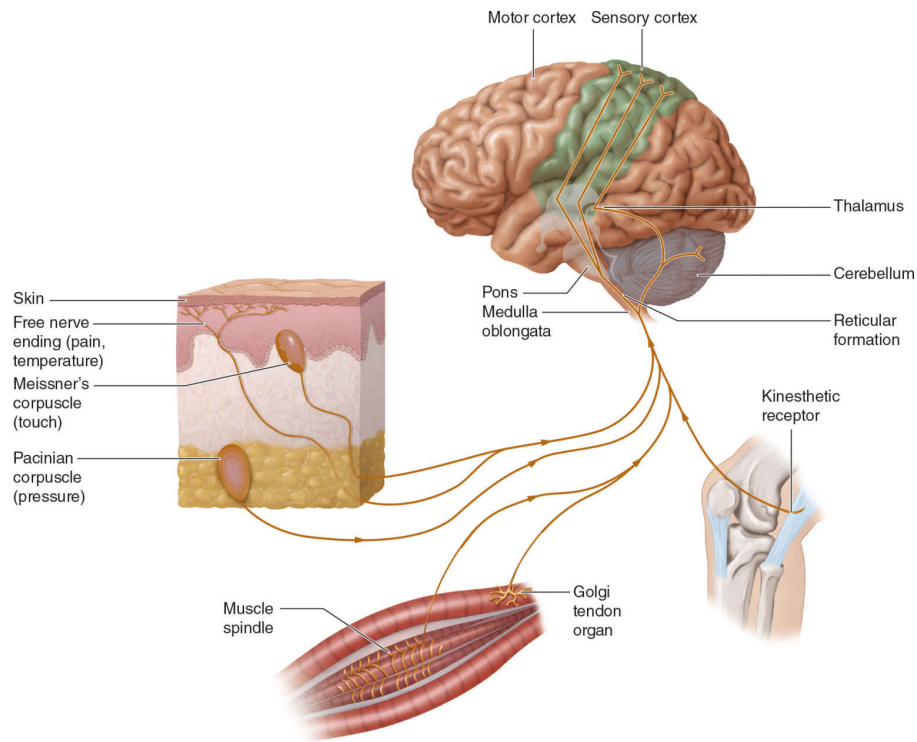


FIGURE 3.8 The sensory receptors and their afferent pathways back to the spinal cord and brain.



- Sensory signals that terminate at the thalamus begin to enter the level of consciousness, and the person begins to distinguish various sensations.
- Only when sensory signals enter the cerebral cortex can one discretely localize the signal. The primary sensory cortex, located in the postcentral gyrus (in the parietal lobe), receives general sensory input from receptors in the skin and from proprioceptors in the muscles, tendons, and joints. This area has a map of the body. Stimulation in a specific area of the body is recognized, and its exact location is known instantly. Thus, this part of the conscious brain allows us to be constantly aware of our surroundings and our relationship to them.

Once a sensory impulse is received, it may evoke a motor response, regardless of the level at which the sensory impulse stops. This response can originate from any of three levels:

- The spinal cord
- The lower regions of the brain
- The motor area of the cerebral cortex

As the level of control moves from the spinal cord to the motor cortex, the degree of movement complexity increases from simple reflex control to complicated movements requiring basic thought processes. Motor responses for more complex movement patterns typically originate in the motor cortex of the brain, and the basal ganglia and cerebellum help to coordinate repetitive movements and to smooth out overall movement patterns. Sensory-motor integration may also involve reflex pathways for quick responses and specialized sensory organs within muscles.

Reflex Activity

What happens when one unknowingly puts one's hand on a hot stove? First, the stimuli of heat and pain are received by the thermoreceptors and nociceptors in the hand, and then sensory action potentials travel to the spinal cord, terminating at the level of entry. Once in the spinal cord, these action potentials are integrated instantly by interneurons that connect the sensory and motor neurons. The action potentials move to the motor neurons and travel to the effectors, the muscles controlling the withdrawal of the hand. The result is that the person reflexively withdraws the hand from the hot stove without giving the action any thought.

A **motor reflex** is a preprogrammed response; any time the sensory nerves transmit certain action potentials, the body responds instantly and identically. In examples like the one just used, whether one touches something that is too hot or too cold, thermoreceptors will elicit a reflex to withdraw the hand. Whether the pain arises from heat or from a sharp object, the nociceptors will also cause a withdrawal reflex. By the time one is consciously aware of the specific stimulus (after sensory action potentials also have been transmitted to the primary sensory cortex), the reflex activity is well

under way, if not completed. All neural activity occurs extremely rapidly, but a reflex is the fastest mode of response because the impulse is not transmitted up the spinal cord to the brain before an action occurs. Only one response is possible; no options need to be considered.

RESEARCH PERSPECTIVE 3.3

Sex Differences in Skeletal Muscle Fiber Types

As discussed in this chapter, and in [chapter 1](#), skeletal muscle is made up of different types of fibers that vary in terms of their structure, biochemistry, and function. The fiber-type composition of different skeletal muscles depends in part on the anatomical location and function of the muscle. However, relatively little is known about whether the proportion of the different fiber types within a skeletal muscle differs between men and women. To date, the few studies that have assessed differential fiber-type composition between sexes have been conducted in rats and mice. In studies that examined sex differences in humans, the fibers measured in men had significantly larger cross-sectional areas, which is not surprising because men have an overall greater muscle mass. However, it appears that women have more type I fibers and fewer type II fibers than their male counterparts on average. When fiber-type composition was examined in the vastus lateralis muscle of a group of men, the average fiber type percentages were 34% type I, 46% type IIa, and 20% type IIx. In women, the percentages were 41% type I, 36% type IIa, and 23% type IIx. This greater prevalence of slow-twitch fibers in women corresponds with a lower contractile velocity in women compared to men but allows for increased endurance and recovery in women.⁸ These data highlight sex differences in muscle fiber-type composition beyond that associated with muscle size alone. This has important implications. Future studies examining skeletal muscle composition, function, and adaptive responses to different forms of exercise training, as well as in pathophysiological conditions, should consider potential sex differences.

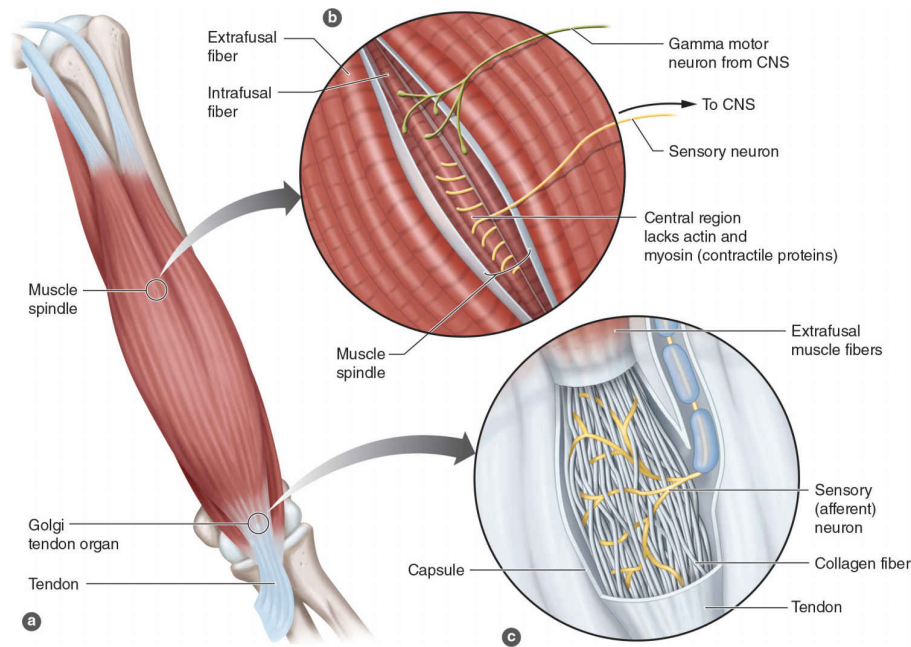


FIGURE 3.9 (a) A muscle belly showing (b) a muscle spindle and (c) a Golgi tendon organ.



Muscle Spindles

Now that we have covered the basics of reflex activity, we can look more closely at two specific reflexes that help control muscle function. The first involves a special structure: the muscle spindle (see [figure 3.9](#)).

The **muscle spindle** is a group of specialized muscle fibers found between regular skeletal muscle fibers, referred to as *extrafusal* (outside the spindle) fibers. A muscle spindle consists of 4 to 20 small, specialized *intrafusal* (inside the spindle) fibers and the nerve endings, sensory and motor, associated with these fibers. A connective tissue sheath surrounds the muscle spindle and attaches to the endomysium of the extrafusal fibers. The intrafusal fibers are controlled by specialized motor neurons, referred to as *γ -motor neurons* (or *gamma motor neurons*). In contrast, extrafusal fibers (the regular fibers) are controlled by *α -motor neurons*.

The central region of an intrafusal fiber cannot contract because it contains no or only a few actin and myosin filaments. This central region can only stretch. Because the muscle spindle is attached to the extrafusal fibers, any time those fibers are stretched, the central region of the muscle spindle is also stretched.

Sensory nerve endings wrapped around this central region of the muscle spindle transmit information to the spinal cord when this region is stretched, transmitting a signal to the CNS about the muscle's length. In the spinal cord, the sensory neuron synapses with an α -motor neuron, which triggers reflexive muscle contraction (in the extrafusal fibers) to resist further stretching.

Let's illustrate this action with an example. A person's arm is bent at the elbow, and the hand is extended, palm up. Suddenly someone places a heavy weight in the palm. The forearm starts to drop, which stretches the muscle fibers in the elbow flexors (e.g., biceps brachii), which in turn stretch the muscle spindles. In response to that stretch, the sensory neurons send action potentials to the spinal cord, which then activates the α -motor neurons of motor units in the same muscles. This activation causes the muscles to increase their force production, overcoming the stretch.

γ -Motor neurons excite the intrafusal fibers, prestretching them slightly. Although the midsection of the intrafusal fibers cannot contract, the ends can. The γ -motor neurons cause slight contraction of the ends of these fibers, which stretches the central region slightly. This prestretch makes the muscle spindle highly sensitive to even small degrees of stretch.

The muscle spindle also assists normal muscle action. It appears that when the α -motor neurons are stimulated to contract the extrafusal muscle fibers, the γ -motor neurons are also activated, contracting the ends of the intrafusal fibers. This stretches the central region of the muscle spindle, giving rise to sensory impulses that travel to the spinal cord and then to the α -motor neurons. In response, the muscle increases its force production. Thus, muscle force production is enhanced through this function of the muscle spindles.

Information brought into the spinal cord from the sensory neurons associated with muscle spindles does not merely end at that level.

Impulses are also sent up to higher parts of the CNS, supplying the brain with continuous feedback on the exact length of the muscle and the rate at which that length is changing. This information is essential for maintaining muscle tone and posture and for executing movements. The muscle spindle functions as a servomechanism to continuously correct movements that do not proceed as planned. The brain is informed of errors in the intended movement at the same time that the error is being corrected at the spinal cord level.



Golgi Tendon Organs

Golgi tendon organs are encapsulated sensory receptors through which a small bundle of muscle tendon fibers pass. These organs

are located just proximal to the tendon fibers' attachment to the muscle fibers, as shown in [figure 3.9](#). Approximately 5 to 25 muscle fibers are usually connected with each Golgi tendon organ. Whereas muscle spindles monitor the length of a muscle, Golgi tendon organs are sensitive to tension in the muscle–tendon complex and operate like a strain gauge, a device that senses changes in tension. Their sensitivity is so great that they can respond to the contraction of a single muscle fiber. These sensory receptors are inhibitory in nature, performing a protective function by reducing the potential for injury. When stimulated, these receptors inhibit the contracting (agonist) muscles and excite the antagonist muscles.

In Review

- Sensory-motor integration is the process by which the PNS relays sensory input to the CNS and the CNS interprets this information and then sends out the appropriate motor signal to elicit the desired motor response.
- The level of nervous system response to sensory input varies according to the complexity of movement necessary. Most simple reflexes are handled by the spinal cord, whereas complex reactions and movements require activation of higher centers in the brain.
- Sensory input can terminate at various levels of the CNS. Not all of this information reaches the brain.
- Reflexes are the simplest form of motor control. These are not conscious responses. For a given sensory stimulus, the motor response is always identical and instantaneous.
- Muscle spindles trigger reflexive muscle action when stretched.
- Golgi tendon organs trigger a reflex that inhibits contraction if the tendon fibers are stretched from high muscle tension.

RESEARCH PERSPECTIVE 3.4

Nontraditional Factors That Impair Neuromuscular Control

Lower-extremity musculoskeletal injuries that occur during sport and physical activity, such as anterior cruciate ligament (ACL) tears, are far too common and extremely costly. Furthermore, these injuries are associated with serious long-term consequences beyond the injury itself, including the accelerated

development of osteoarthritis. The first step toward effectively preventing lower-extremity injuries is the appropriate identification of the important risk factors for injury.

Perhaps the most commonly considered primary injury risk factors are measures of neuromuscular control, such as balance and movement technique. However, beyond these traditional risk factors, it is important to consider nontraditional factors that may predispose the athlete to injury, such as alterations in hydration, increases in body temperature, and fatigue. Importantly, hypohydration (below-optimal body fluid balance), hyperthermia (increased body core temperature), and fatigue, which are likely to be encountered during physical activity, all impair neuromuscular control. A 2012 study has substantiated this finding.³ In particular, hypohydration combined with hyperthermia negatively affected movement technique and, to a lesser extent, balance. These findings emphasize the need for adequate hydration during exercise, especially when performed in hot environments, not only to optimize performance and prevent heat-related complications (see [chapter 12](#)), but also to reduce the risk of lower-extremity injury.¹

Golgi tendon organs are important in resistance exercise. They function as safety devices, helping to prevent the muscle from developing excessive force during a contraction that may ultimately damage the muscle. Additionally, some researchers speculate that reducing the influence of Golgi tendon organs disinhibits the active muscles, allowing a more forceful muscle action. This mechanism may explain at least part of the gains in muscular strength that accompany strength training.

Motor Response

Now that we have discussed how sensory input is integrated to determine the appropriate motor response, the last step in the process is how muscles respond to motor action potentials once they reach the muscle fibers.

Once an action potential reaches an α -motor neuron, it travels the length of the neuron to the NMJ. From there, the action potential spreads to all muscle fibers innervated by that particular α -motor neuron. Recall that the α -motor neuron and all muscle fibers it innervates form a single motor unit. Each muscle fiber is innervated by only one α -motor neuron, but each α -motor neuron innervates up to several thousand muscle fibers, depending on the function of the muscle. Muscles controlling fine movements have only a small

number of muscle fibers per α -motor neuron. The muscles that control eye movements (the extraocular muscles) have an innervation ratio of 1:15, meaning that one α -motor neuron controls only 15 muscle fibers. Muscles with more general functions have many fibers per α -motor neuron. For example, the gastrocnemius and tibialis anterior muscles of the lower leg have innervation ratios of almost 1:2,000.

The muscle fibers in a specific motor unit are homogeneous with respect to fiber type. Thus, one will not find a motor unit that has both type II and type I fibers. In fact, as mentioned in [chapter 1](#), it is generally believed that the characteristics of the α -motor neuron actually determine the fiber type in the given motor unit.⁷

IN CLOSING

In this chapter, we examined how the nervous system is organized and how that organization functions to control movement. We covered the central nervous system as it relates to movement and the sensory and effector arms of the peripheral nervous system. We have seen how muscles respond to neural stimulation, whether through reflexes or under complex control of the higher brain centers, and the role of individual motor units in determining this response. Thus, we have learned how the body functions to allow people to move. In the next chapter, we examine the role of hormones in the body's response to exercise.

KEY TERMS

acetylcholine
adrenergic
axon hillock
axon terminal
central nervous system (CNS)
cholinergic
depolarization
effector (efferent) nerves
end branches
excitatory postsynaptic potential (EPSP)
Golgi tendon organ
graded potential
hyperpolarization
inhibitory postsynaptic potential (IPSP)
motor neurons (motor nerves)
motor reflex
muscle spindle
myelin sheath
nerve impulse
neuromuscular junction
neuron
neurotransmitter
norepinephrine
peripheral nervous system (PNS)
resting membrane potential (RMP)
saltatory conduction
sensory (afferent) nerves

sensory-motor integration
sodium–potassium pump
synapse
threshold

STUDY QUESTIONS

1. What are the major divisions of the nervous system? What are their major functions?
2. Name the different anatomical parts of a neuron, and discuss their function.
3. Explain the resting membrane potential. What causes it? How is it maintained?
4. Describe an action potential. What is required before an action potential is activated?
5. Explain how an action potential is transmitted from a presynaptic neuron to a postsynaptic neuron. Describe a synapse and a neuromuscular junction.
6. What brain centers have major roles in controlling movement, and what are these roles?
7. How do the sympathetic and parasympathetic systems differ? What is their significance in performing physical activity?
8. Explain how reflex movement occurs in response to touching a hot object.
9. Describe the role of the muscle spindle in controlling muscle contraction.
10. Describe the role of the Golgi tendon organ in controlling muscle contraction.

STUDY GUIDE ACTIVITIES

In addition to the activities listed in the chapter opening outline, two other activities are available in the web study guide, located at

www.HumanKinetics.com/PhysiologyOfSportAndExercise

The **KEY TERMS** activity reviews important terms, and the end-of-chapter **QUIZ** tests your understanding of the material covered in the chapter.

